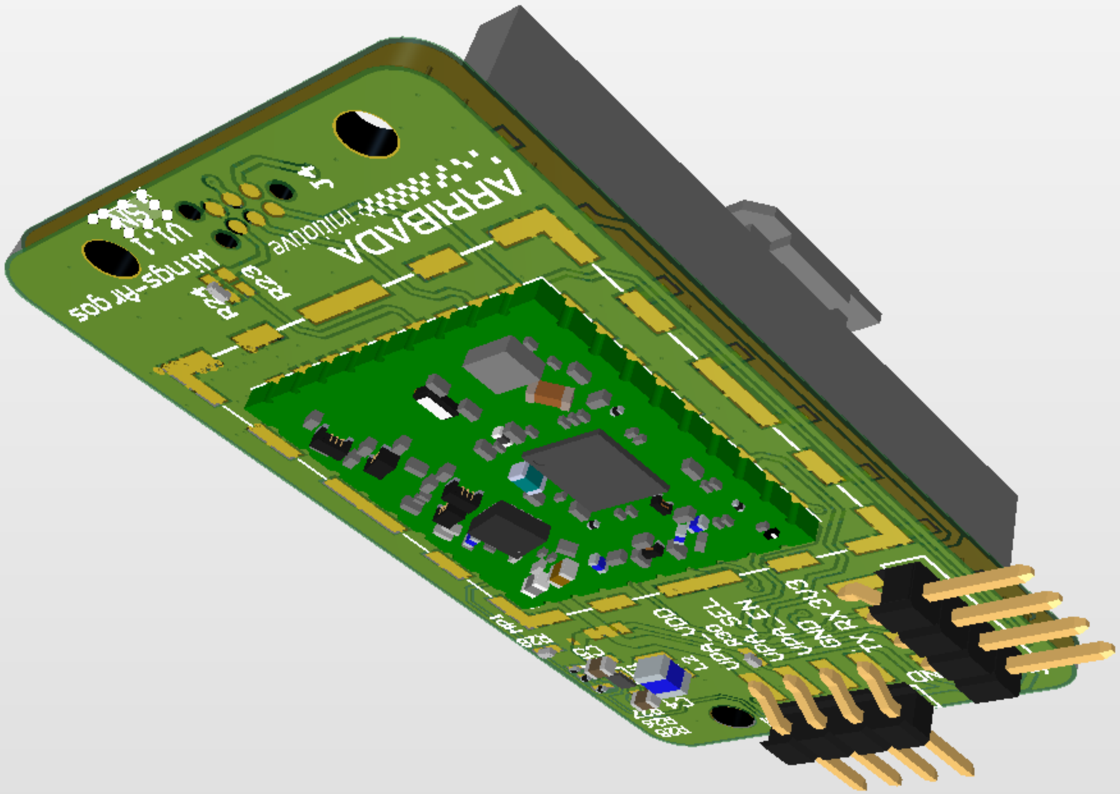

ARRIBADA
initiative

HorizonLink-Wings_Breakout

smd-wings

Feather wing for the ARGOS-SMD module. This breakout board can be used with a Feather board (from Adafruit) to connect the ARGOS-SMD module. It exposes the ARGOS-SMD module pins to the Feather board and includes a specific LDO to power the RF amplifier.

Table of contents

Legend

General comment such as function title, configuration, ...

Text to be added to silkscreen.

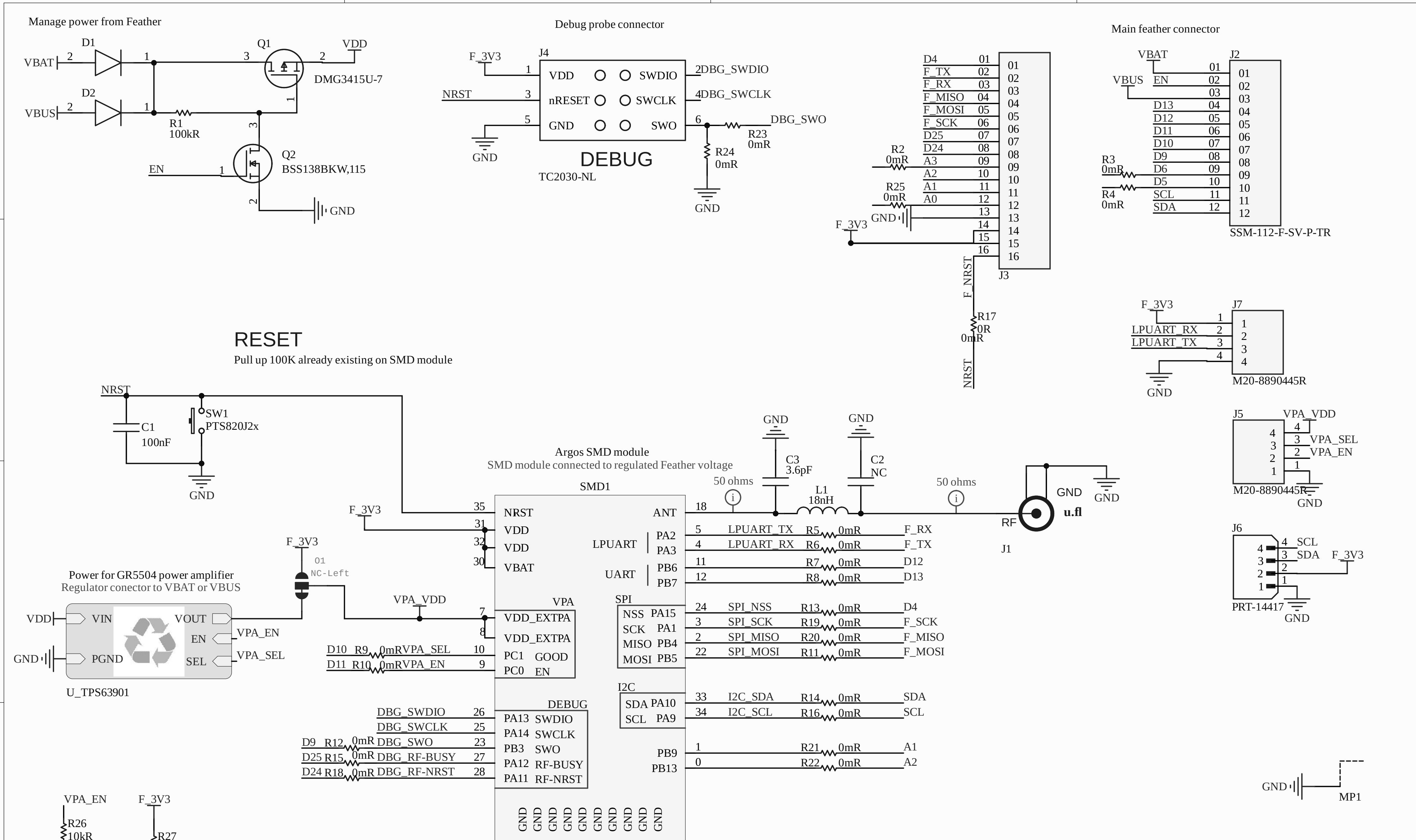
Warning text.

Notes to generate the board layout.

OPEN HARDWARE LICENSE AGREEMENT

smd-wings

Title *HorizonLink-Wings_Breakout*			*Arribada Initiative* *https://github.com/arribada* *contact@arribada.org* *arribada.org* *UK - Portsmouth*
Size: A4	Reference: smd-wings	Revision: A 1.0	
Date: 06/08/2025 Time: 14:39:57		Sheet 1 of 3	
File: C:\Users\fourn\Altium\AD25\Projects\Arribada\featherwings-argos-smd-hw\Project_Overview.SchDoc			



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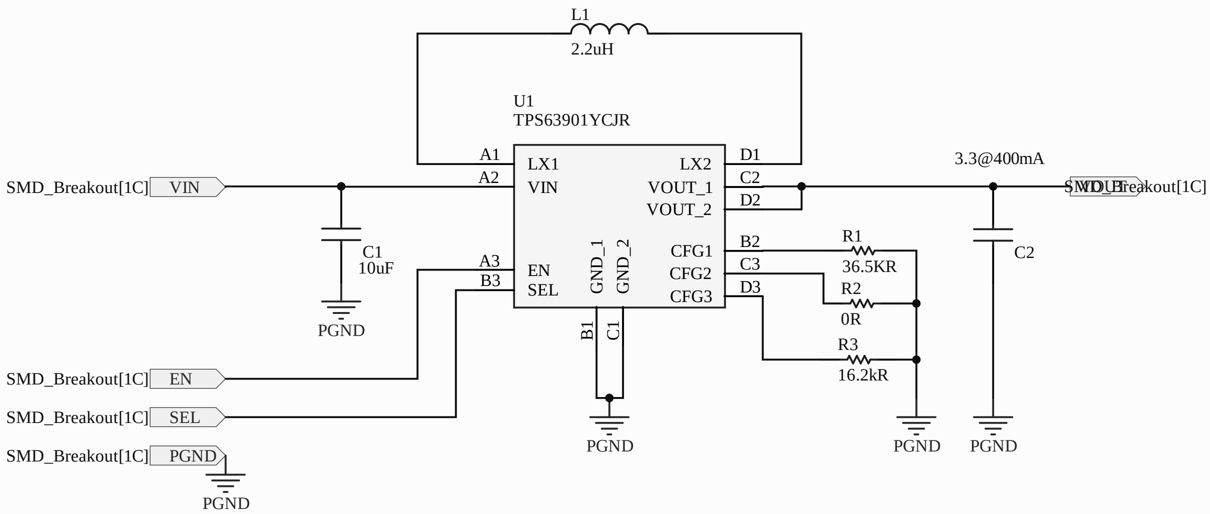
Size: A4	Reference: smd-wings	Revision: A 1.0	
Date: 06/08/2025	Time: 14:39:58	Sheet 2 of 3	
File:	C:\Users\fourn\Altium\AD25\Projects\Arribada\featherwings-argos-smd-hw\SMD_Breakout.SchDoc		

Arribada Initiative
https://github.com/arribada
contact@arribada.org
arribada.org
UK - Portsmouth



Title **HorizonLink-Wings_Breakout**

Size: A4 | Reference: smd-wings | Revision: A 1.0

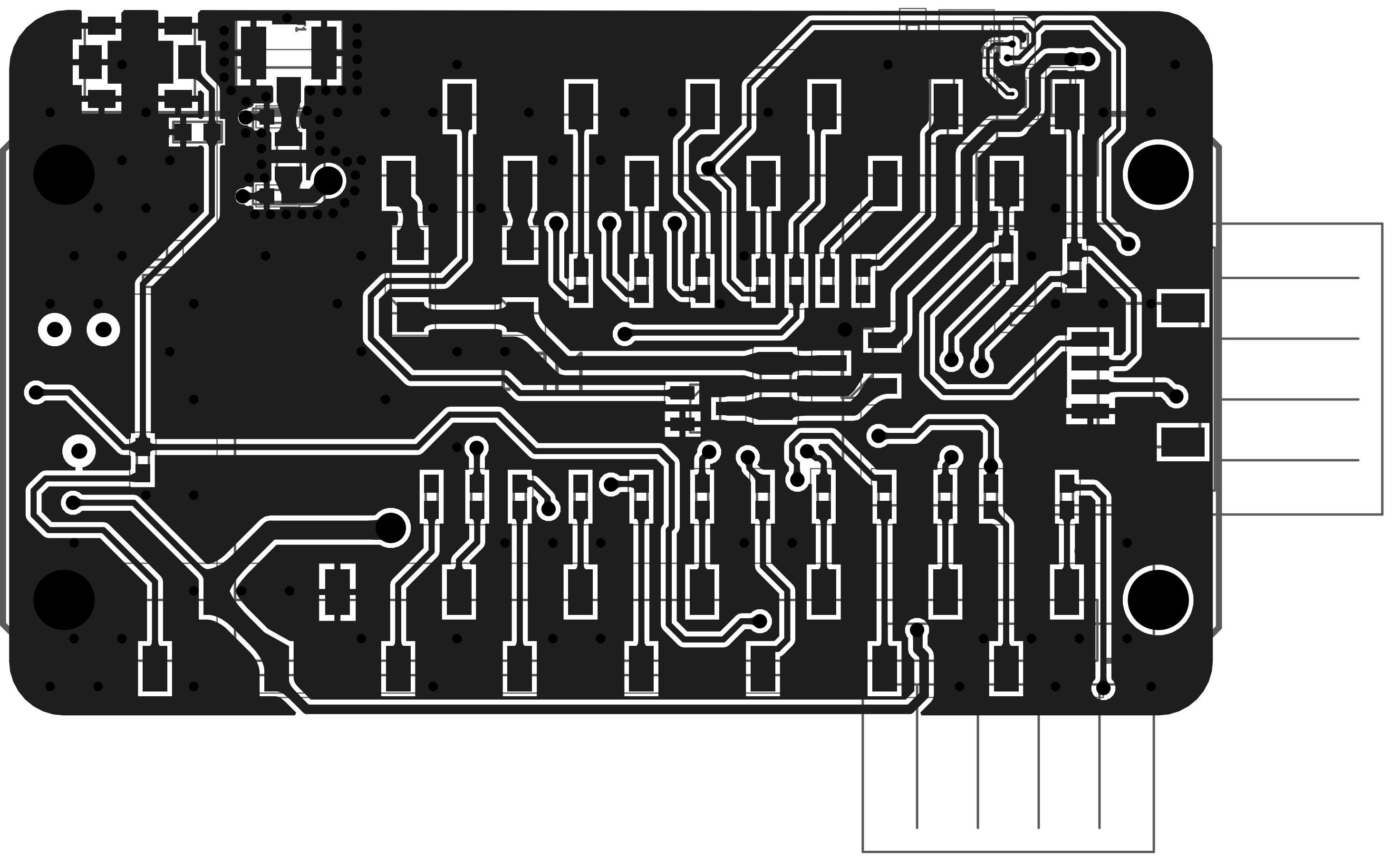
Date: 06/08/2025 | Time: 14:39:58 | Sheet 3 of 3

File: C:\Users\fourn\Altium\AD25\Projects\Arribada\hardware-lib\DeviceSheets\Power\TPS63901_BuckBoost_lowIq_3V3_40

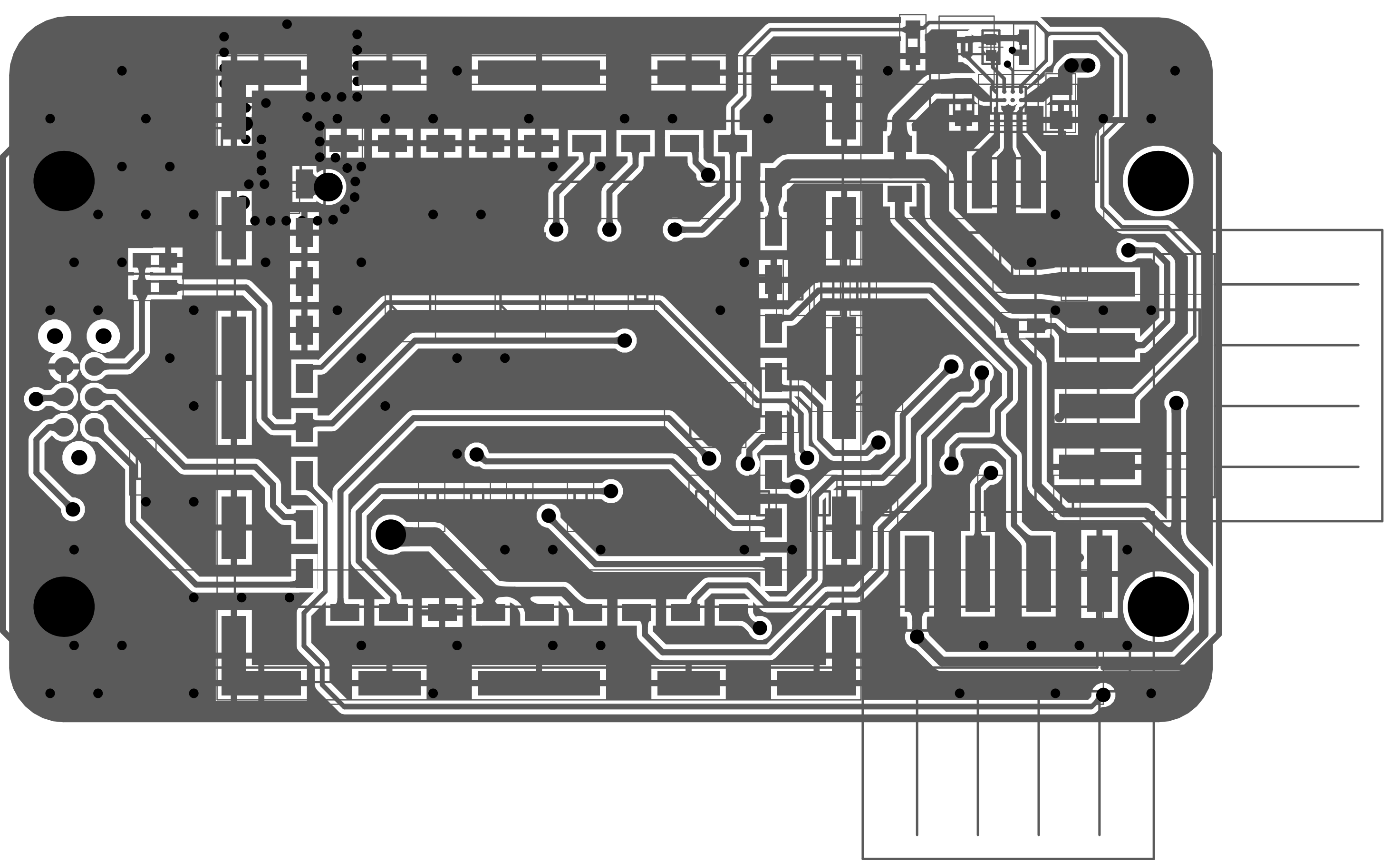
Arribada Initiative
https://github.com/arribada
contact@arribada.org
arribada.org
UK - Portsmouth

Board Stack Report

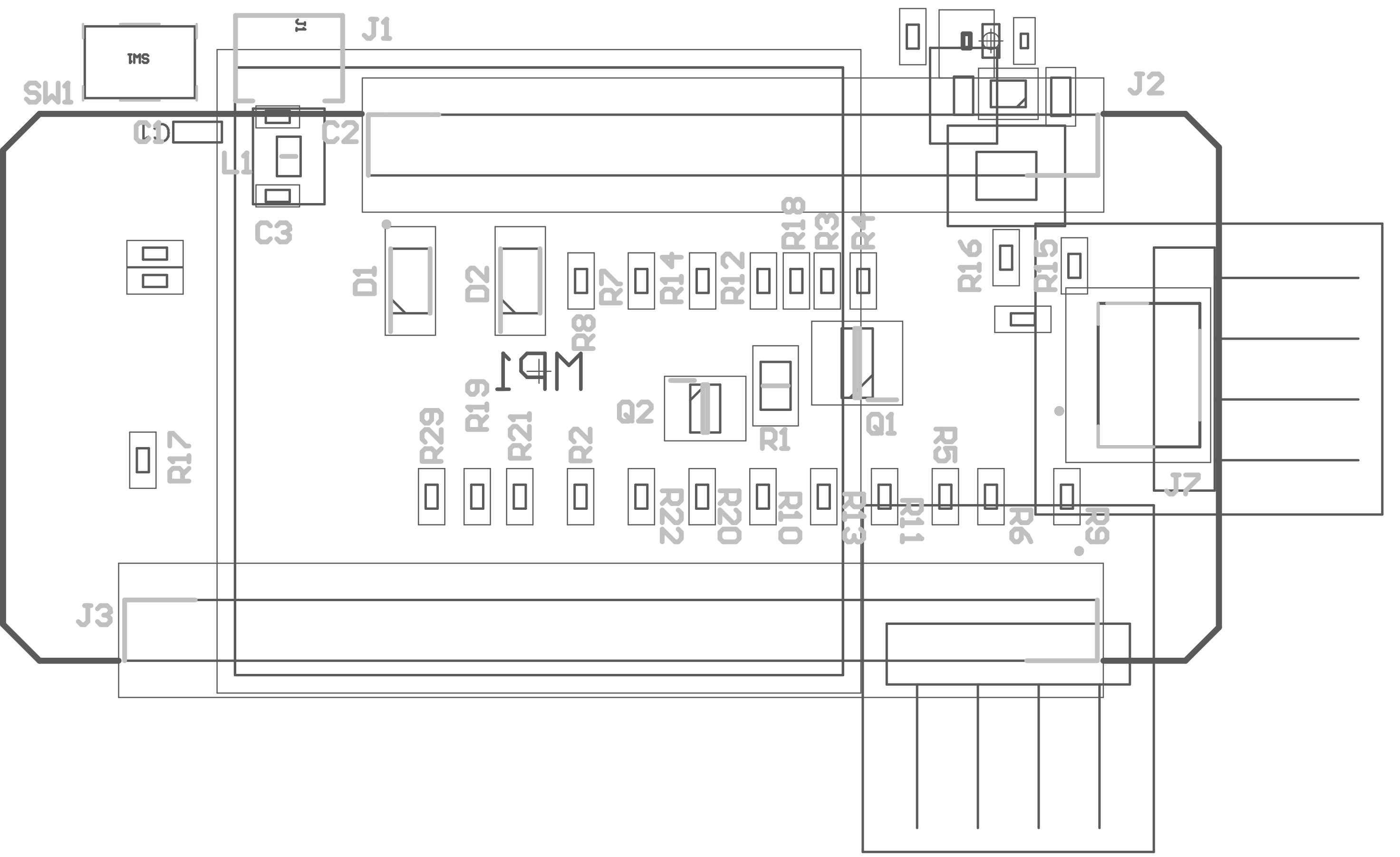
Stack Up		Layer Stack			
Layer	Board Layer Stack	Name	Material	Thickness	Constant
1		Top Overlay		0mil	
2		Top Solder	SM-001	1mil	4
3		Top Surface Finish	PbSn	0.787mil	
4		Top Layer	CF-004	1.378mil	
5		Dielectric 1	Core-043	59mil	4.3
6		Bottom Layer	CF-004	1.378mil	
7		Bottom Surface Finish	PbSn	0.787mil	
8		Bottom Solder	SM-001	1mil	4
9		Bottom Overlay		0mil	
	Height : 65.331mil				



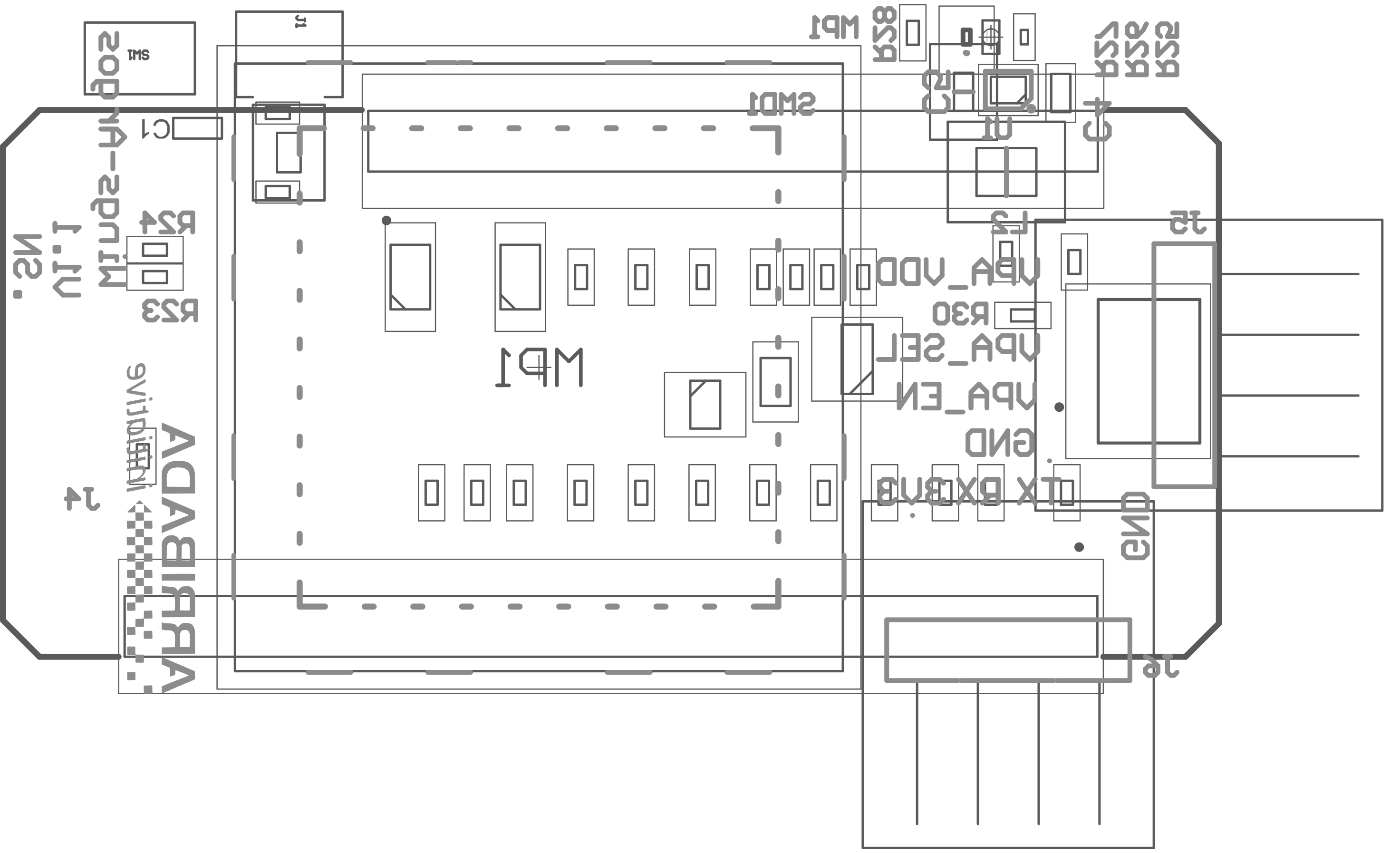
Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	SM-001	1,00mil	4	
	Top Surface Finish	PbSn	0,79mil		
1	Top Layer	CF-004	1,38mil		
	Dielectric 1	Core-043	59,00mil	4.3	
2	Bottom Layer	CF-004	1,38mil		
	Bottom Surface Finish	PbSn	0,79mil		
	Bottom Solder	SM-001	1,00mil	4	
	Bottom Overlay				



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	SM-001	1,00mil	4	
	Top Surface Finish	PbSn	0,79mil		
1	Top Layer	CF-004	1,38mil		
	Dielectric 1	Core-043	59,00mil	4.3	
2	Bottom Layer	CF-004	1,38mil		
	Bottom Surface Finish	PbSn	0,79mil		
	Bottom Solder	SM-001	1,00mil	4	
	Bottom Overlay				



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	SM-001	1,00mil	4	
	Top Surface Finish	PbSn	0,79mil		
1	Top Layer	CF-004	1,38mil		
	Dielectric 1	Core-043	59,00mil	4.3	
2	Bottom Layer	CF-004	1,38mil		
	Bottom Surface Finish	PbSn	0,79mil		
	Bottom Solder	SM-001	1,00mil	4	
	Bottom Overlay				



Layer	Name	Material	Thickness	Constant	Board Layer Stack
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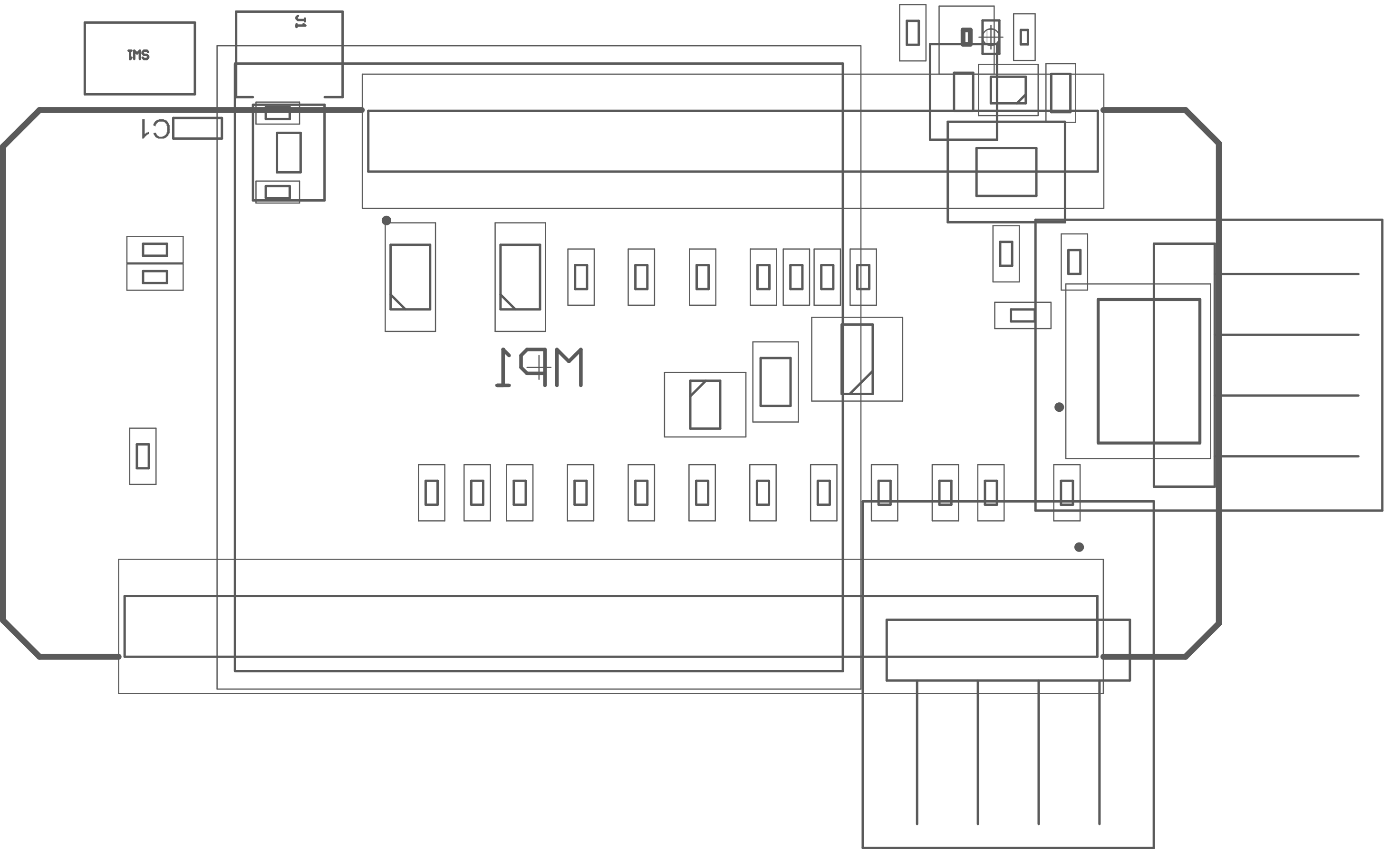
	Top Solder	SM-001	1,00mil	4	
	Top Surface Finish	PbSn	0,79mil		
1	Top Layer	CF-004	1,38mil		
	Dielectric 1	Core-043	59,00mil	4.3	
2	Bottom Layer	CF-004	1,38mil		
	Bottom Surface Finish	PbSn	0,79mil		
	Bottom Solder	SM-001	1,00mil	4	
	Bottom Overlay				



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	SM-001	1,00mil	4	
	Top Surface Finish	PbSn	0,79mil		
1	Top Layer	CF-004	1,38mil		
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2	Bottom Layer	CF-004	1,38mil		
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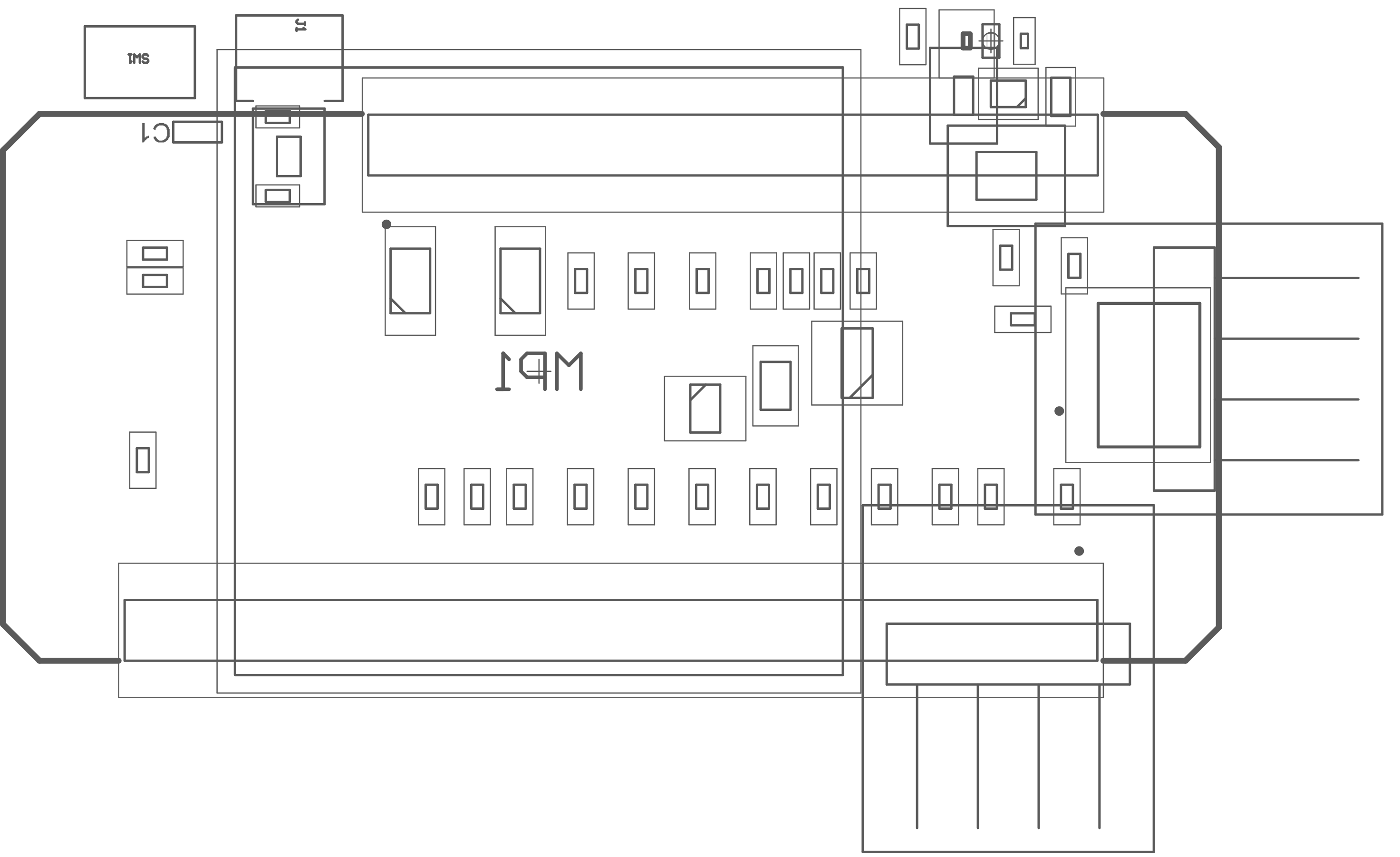
	Bottom Solder	SM-001	1,00mil	4	
	Bottom Overlay				



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	SM-001	1,00mil	4	
	Top Surface Finish	PbSn	0,79mil		
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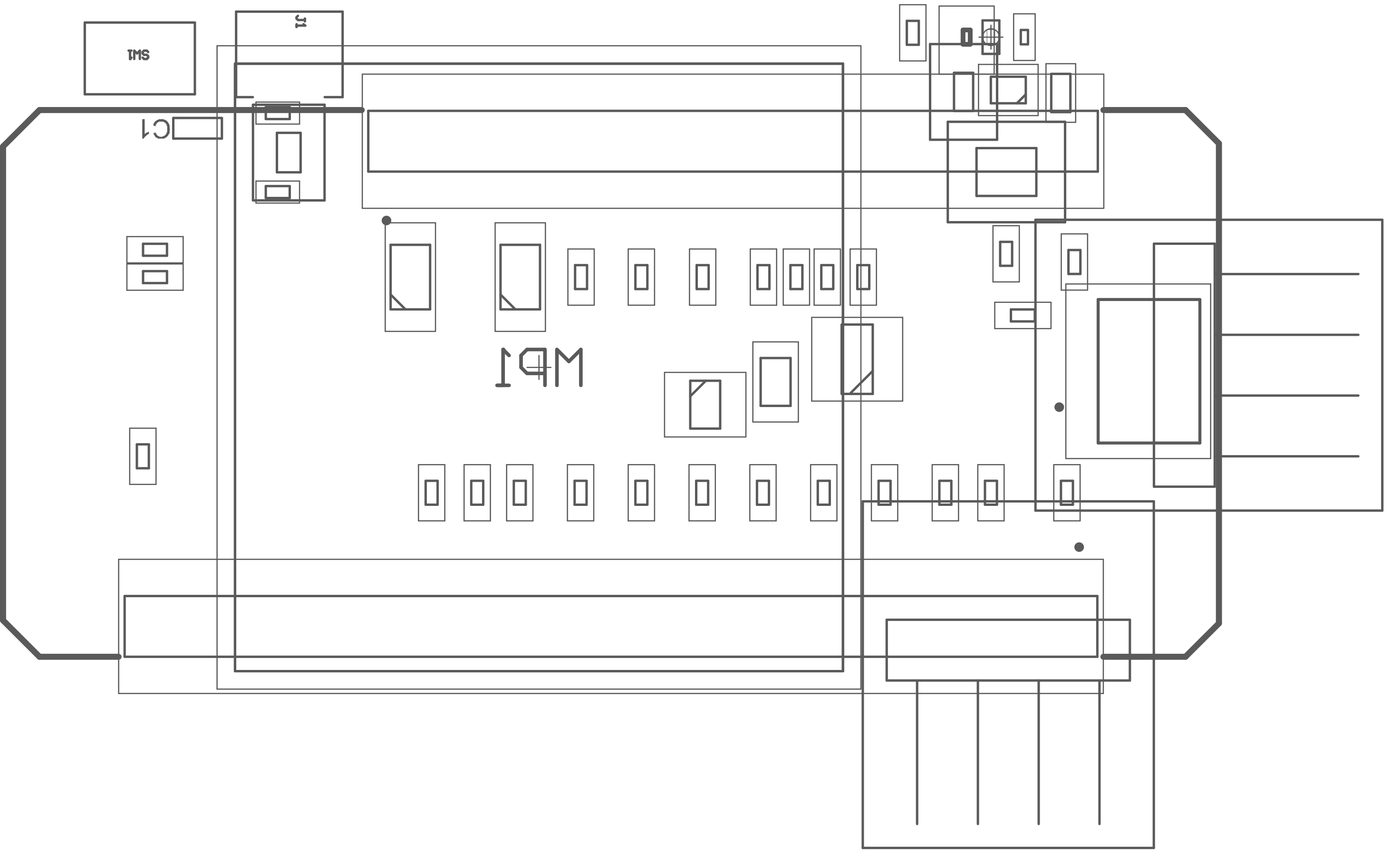
	Bottom Solder	SM-001	1,00mil	4	
	Bottom Overlay				



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	SM-001	1,00mil	4	
	Top Surface Finish	PbSn	0,79mil		
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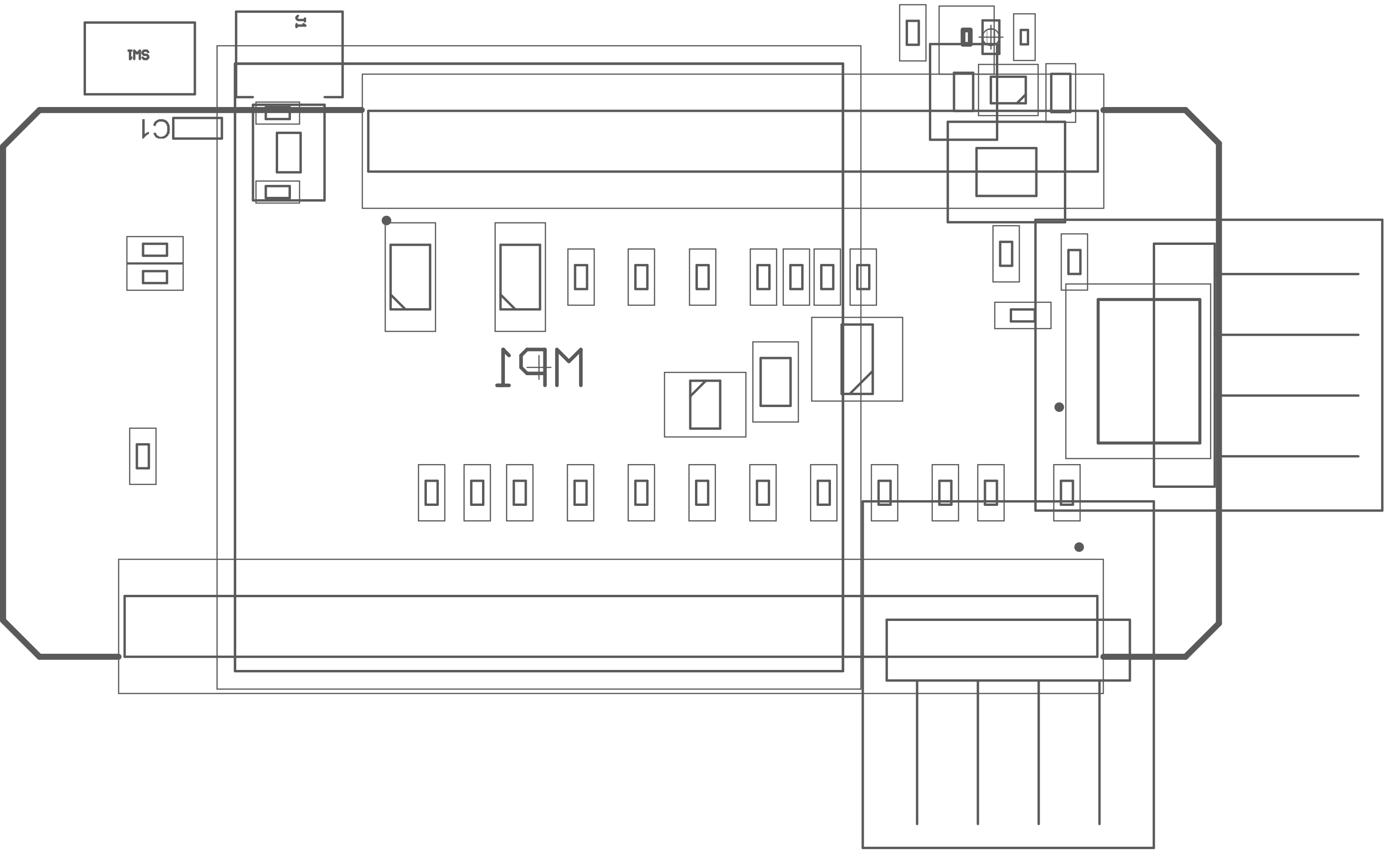
	Bottom Solder	SM-001	1,00mil	4	
	Bottom Overlay				



Layer	Name	Material	Thickness	Constant	Board Layer Stack
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	Top Solder	SM-001	1,00mil	4	
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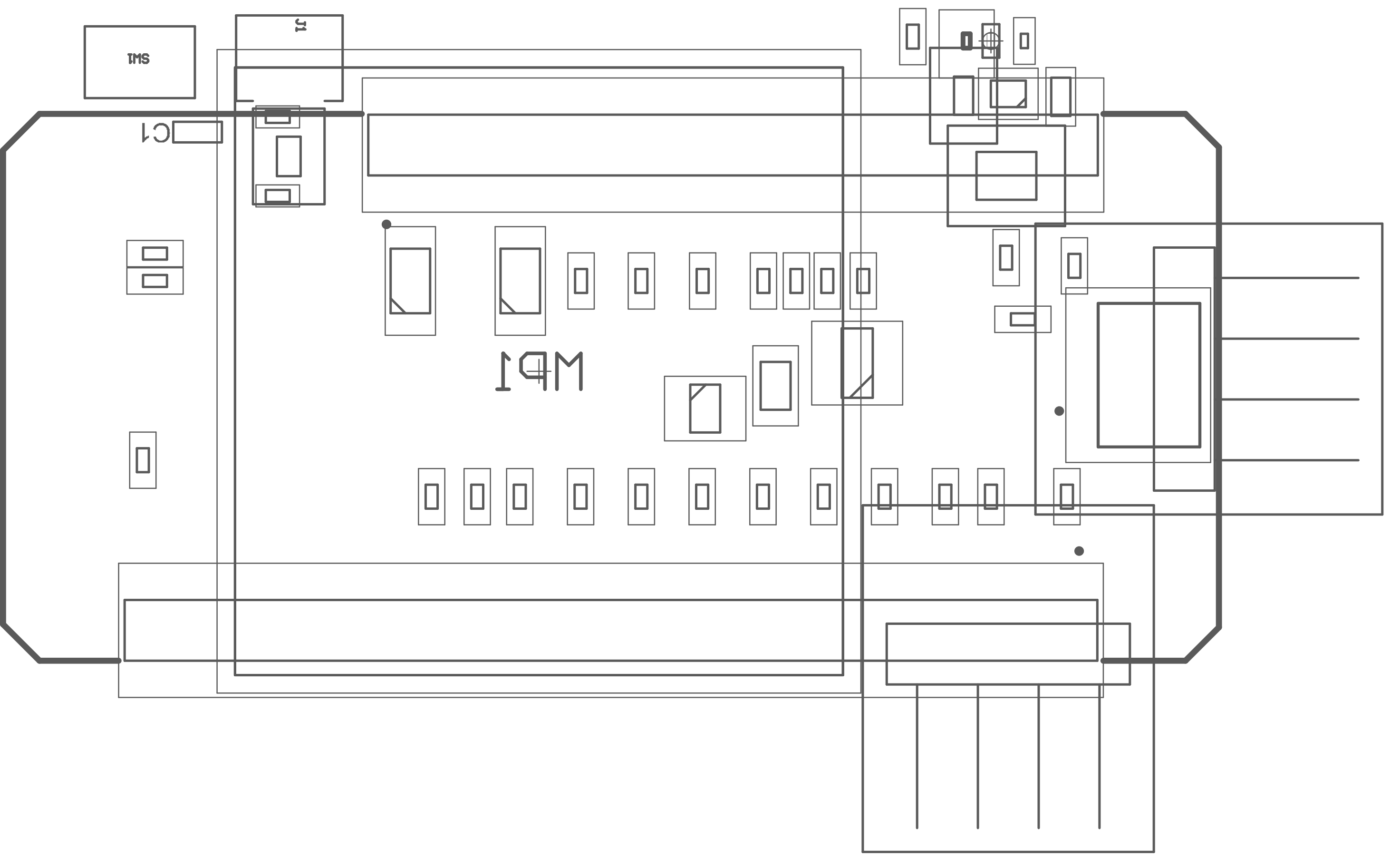
	Bottom Solder	SM-001	1,00mil	4	
	Bottom Overlay				



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	SM-001	1,00mil	4	
	Top Surface Finish	PbSn	0,79mil		
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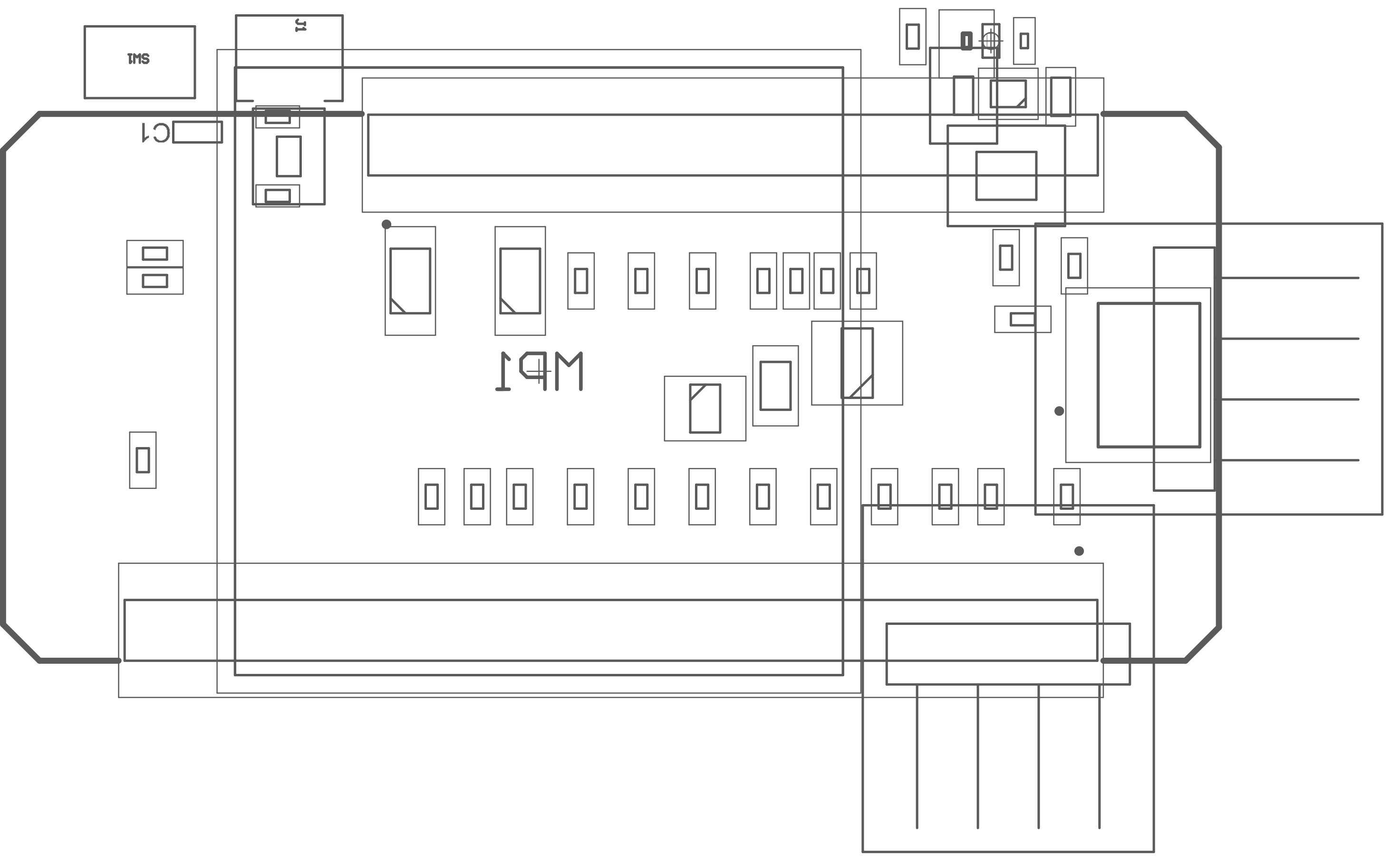
	Bottom Overlay				



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	SM-001	1,00mil	4	
	Top Surface Finish	PbSn	0,79mil		
1	Top Layer	CF-004	1,38mil		
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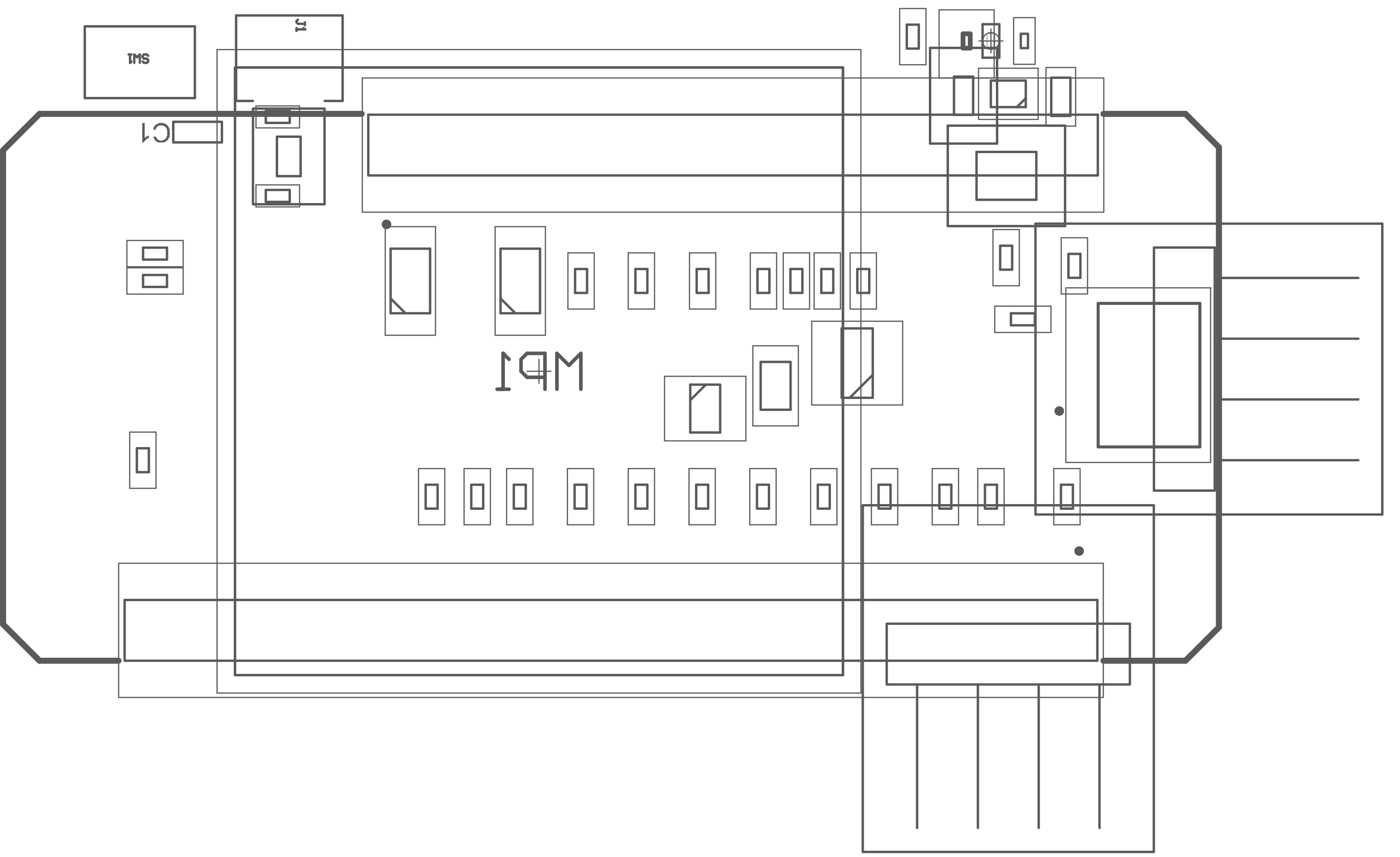
	Bottom Solder	SM-001	1,00mil	4	
	Bottom Overlay				



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
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	Top Surface Finish	PbSn	0,79mil		
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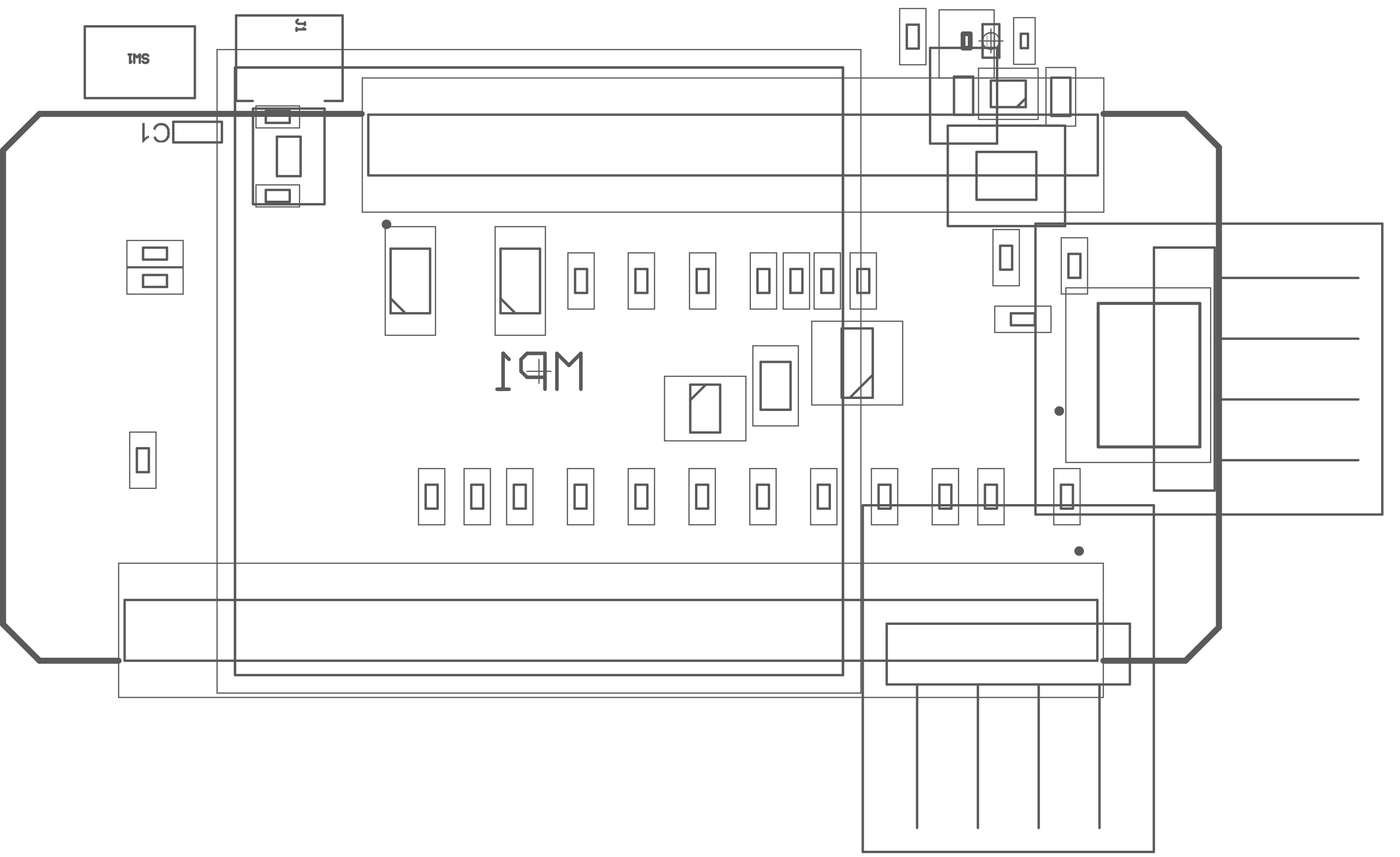
	Bottom Overlay				



Layer	Name	Material	Thickness	Constant	Board Layer Stack
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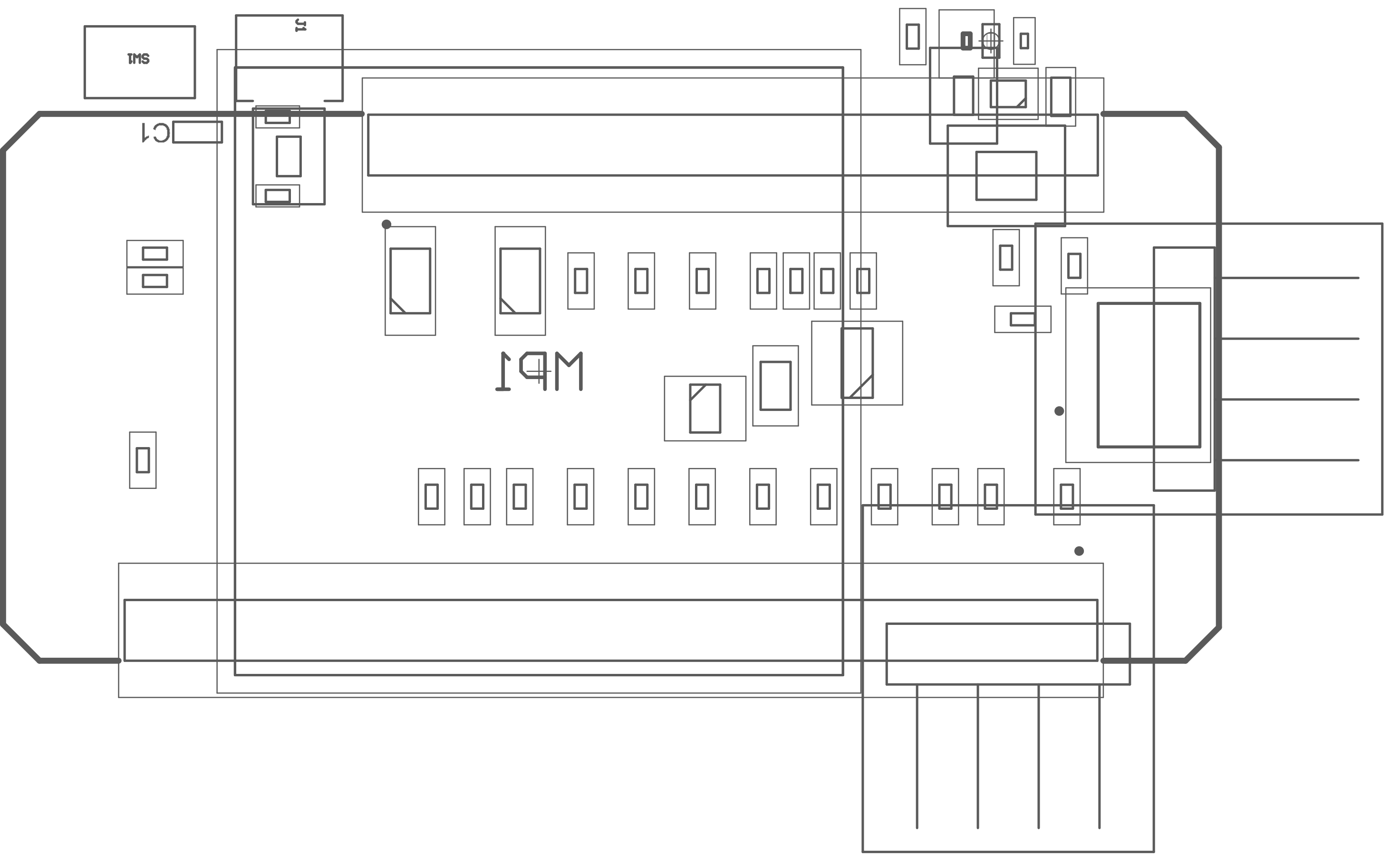
	Bottom Overlay				



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
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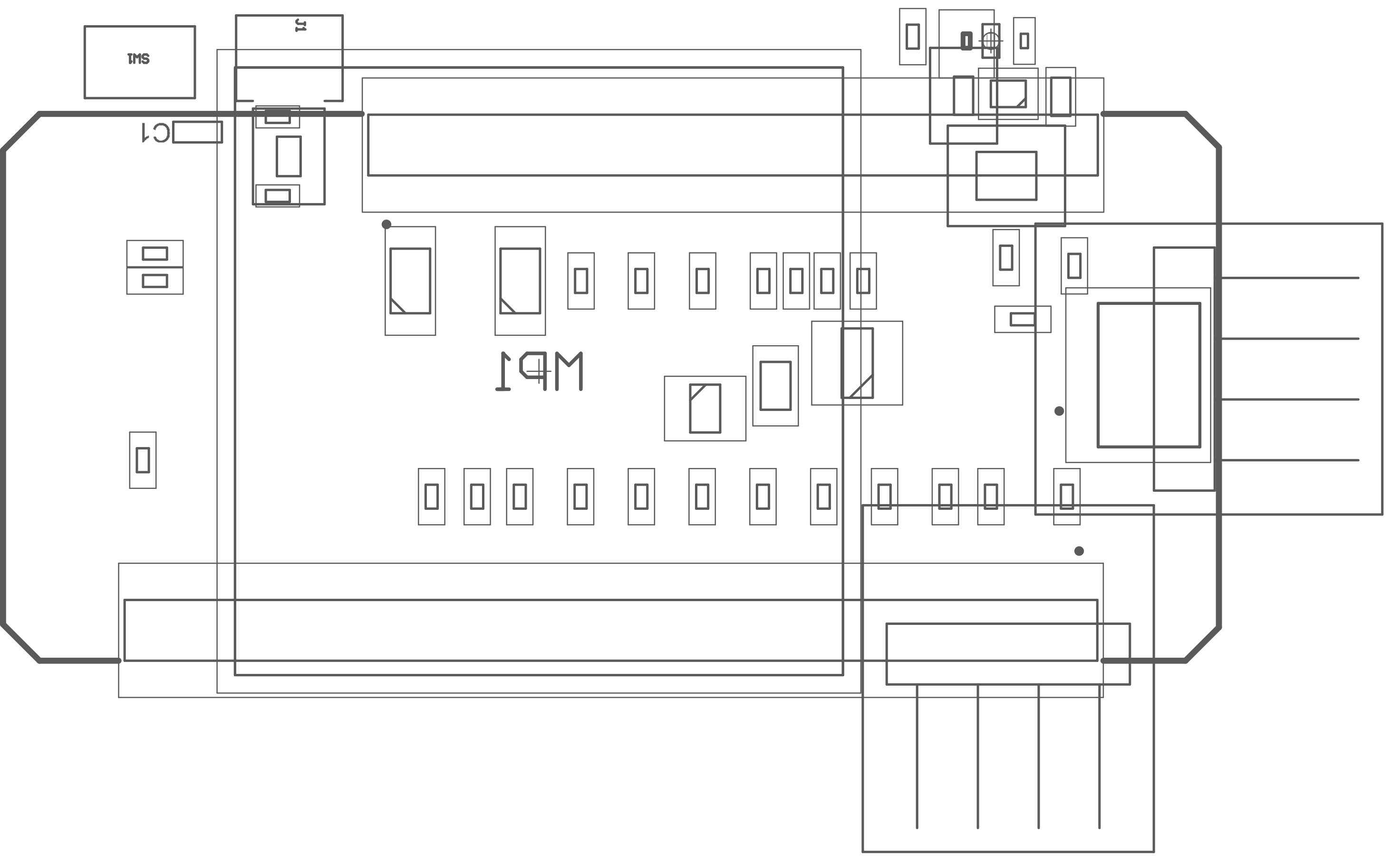
	Bottom Overlay				



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	SM-001	1,00mil	4	
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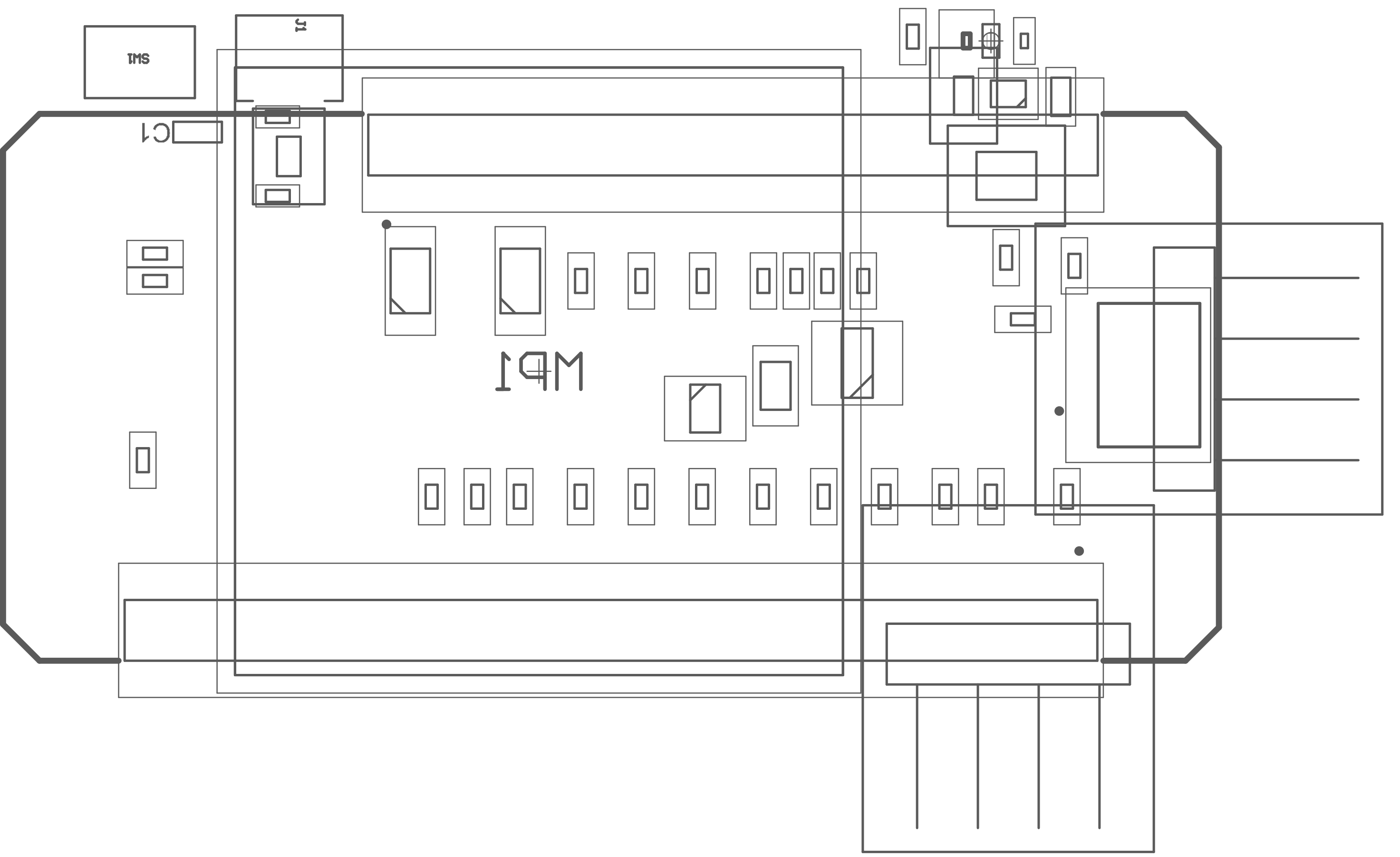

Layer	Name	Material	Thickness	Constant	Board Layer Stack
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	Top Solder	SM-001	1,00mil	4	
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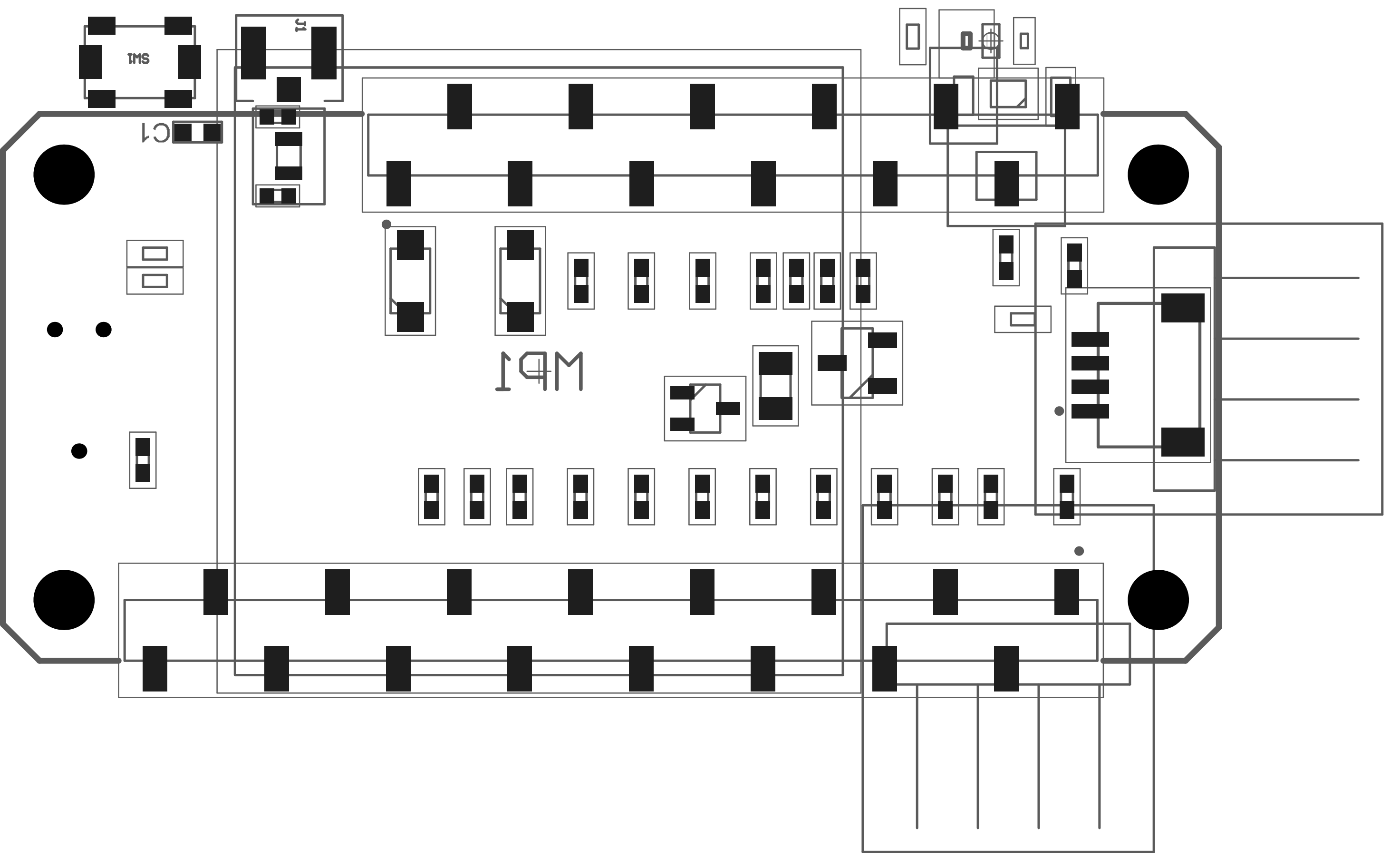
	Bottom Overlay				



Layer	Name	Material	Thickness	Constant	Board Layer Stack
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	Top Surface Finish	PbSn	0,79mil		
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	Bottom Surface Finish	PbSn	0,79mil		
	Bottom Solder	SM-001	1,00mil	4	
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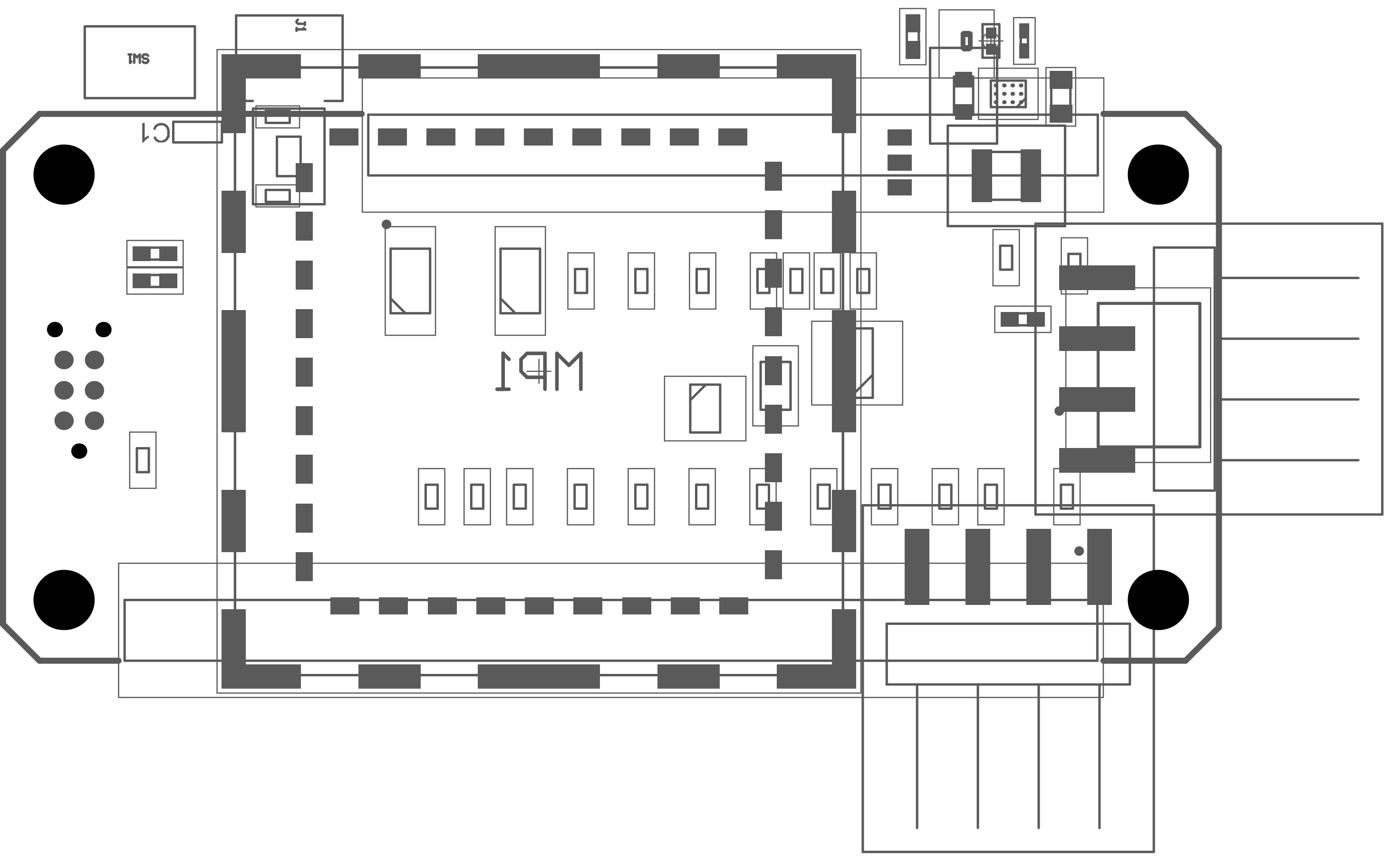

Layer	Name	Material	Thickness	Constant	Board Layer Stack
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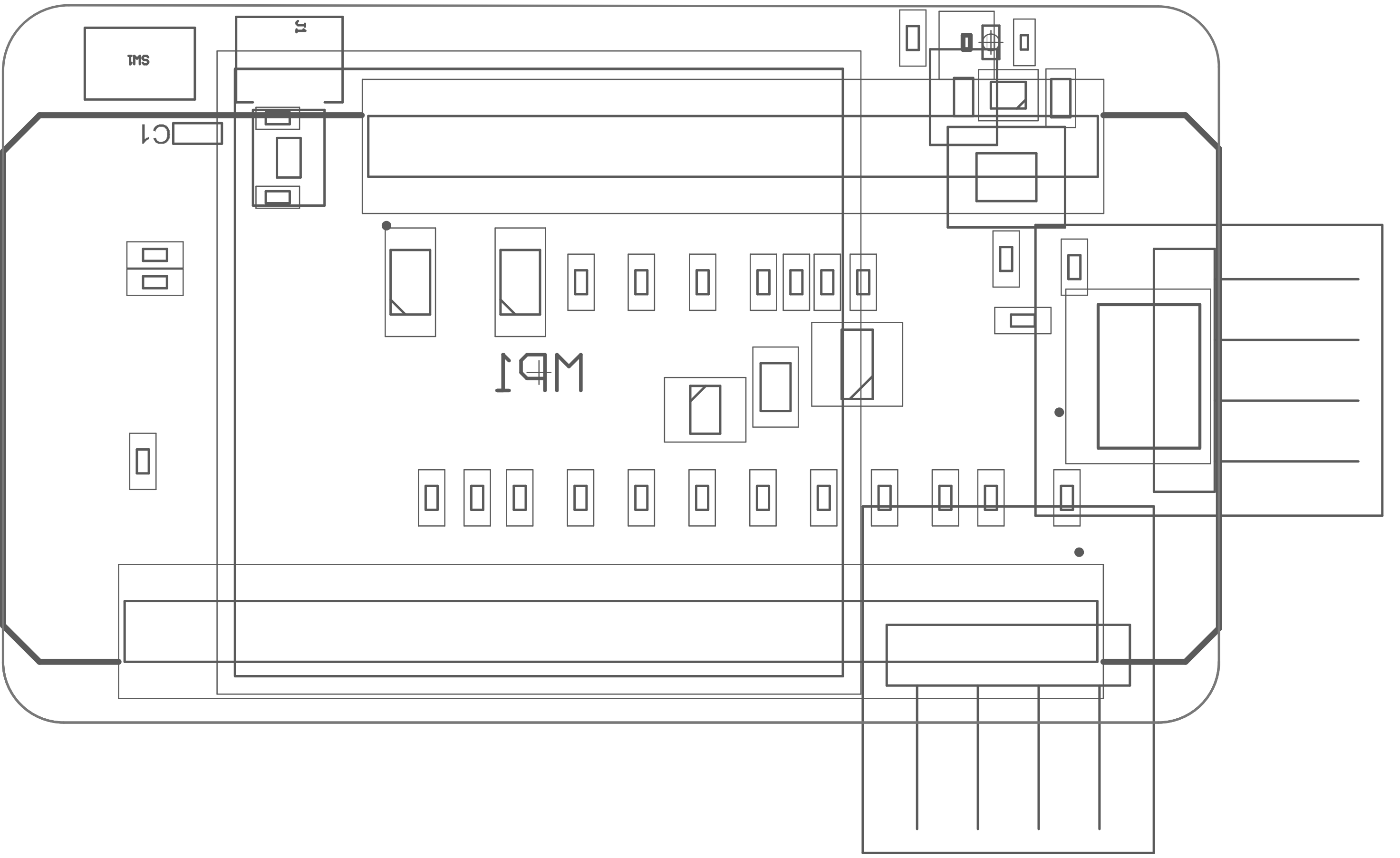

Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
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	Top Surface Finish	PbSn	0,79mil		
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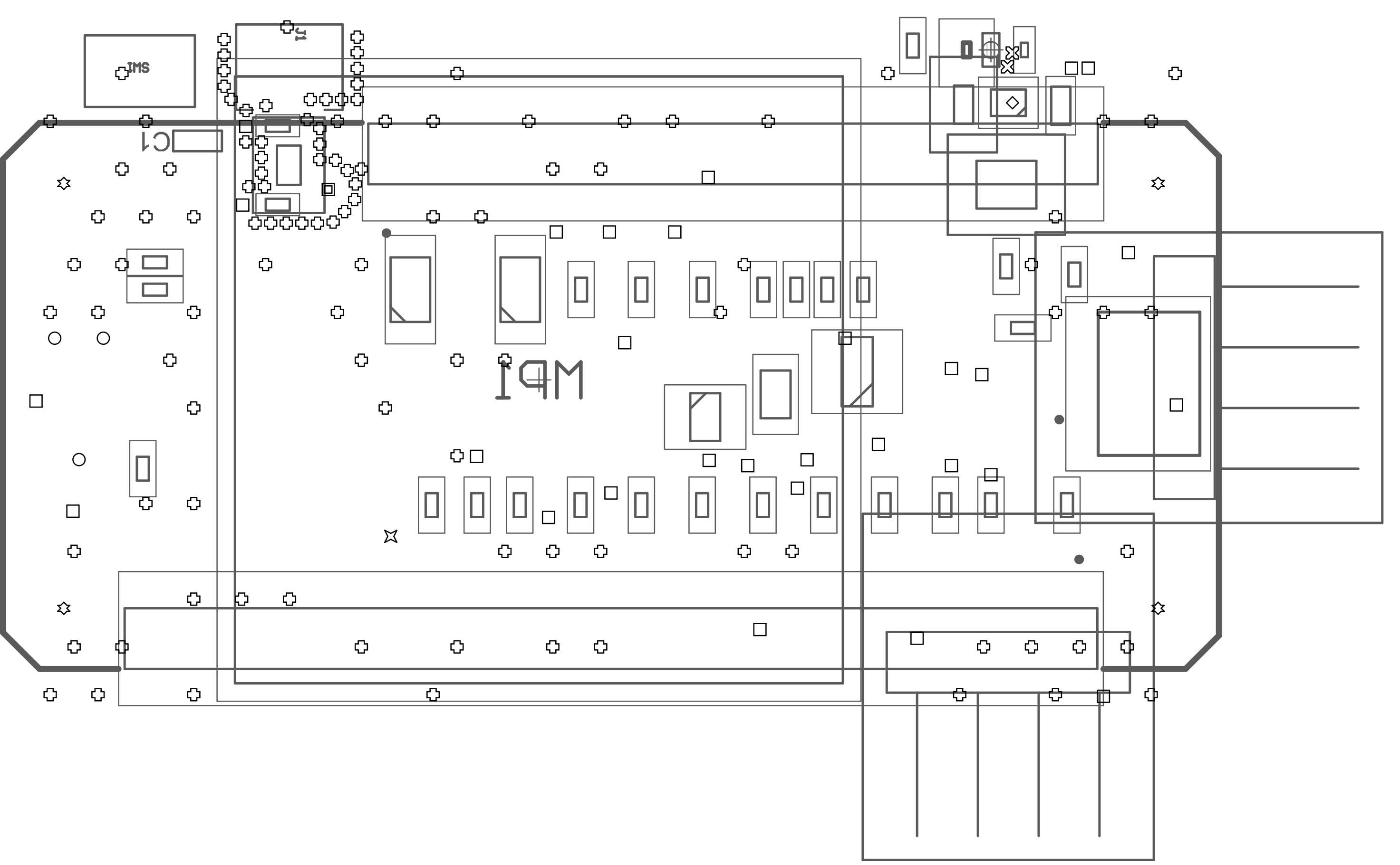

Symbol	Count	Hole Size	Plated	Hole Type	Drill Layer Pair	Via/Pad	Pad Shape	Template	Description	Hole Tolerance (+)	Hole Tolerance (-)
◇	1	0,100mm (3,94mil)	PTH	Round	Top Layer - Bottom Layer	Via	Rounded	v17h10			
▣	1	0,600mm (23,62mil)	PTH	Round	Top Layer - Bottom Layer	Via	Rounded	v120h60m140mx0			
⌑	1	0,711mm (27,99mil)	PTH	Round	Top Layer - Bottom Layer	Via	Rounded	v127h71m147mx0			
⊠	2	0,150mm (5,91mil)	PTH	Round	Top Layer - Bottom Layer	Via	Rounded	v30h15			
○	3	0,991mm (39,02mil)	NPTH	Round	Top Layer - Bottom Layer	Pad	Rounded	c65hn99			
✡	4	2,540mm (100,00mil)	NPTH	Round	Top Layer - Bottom Layer	Pad	Rounded	c254hn254			
□	29	0,300mm (11,81mil)	PTH	Round	Top Layer - Bottom Layer	Via	Rounded	(Mixed)			
✚	113	0,200mm (7,87mil)	PTH	Round	Top Layer - Bottom Layer	Via	Rounded	(Mixed)			
	154 Total										

Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	SM-001	1,00mil	4	
	Top Surface Finish	PbSn	0,79mil		
1	Top Layer	CF-004	1,38mil		
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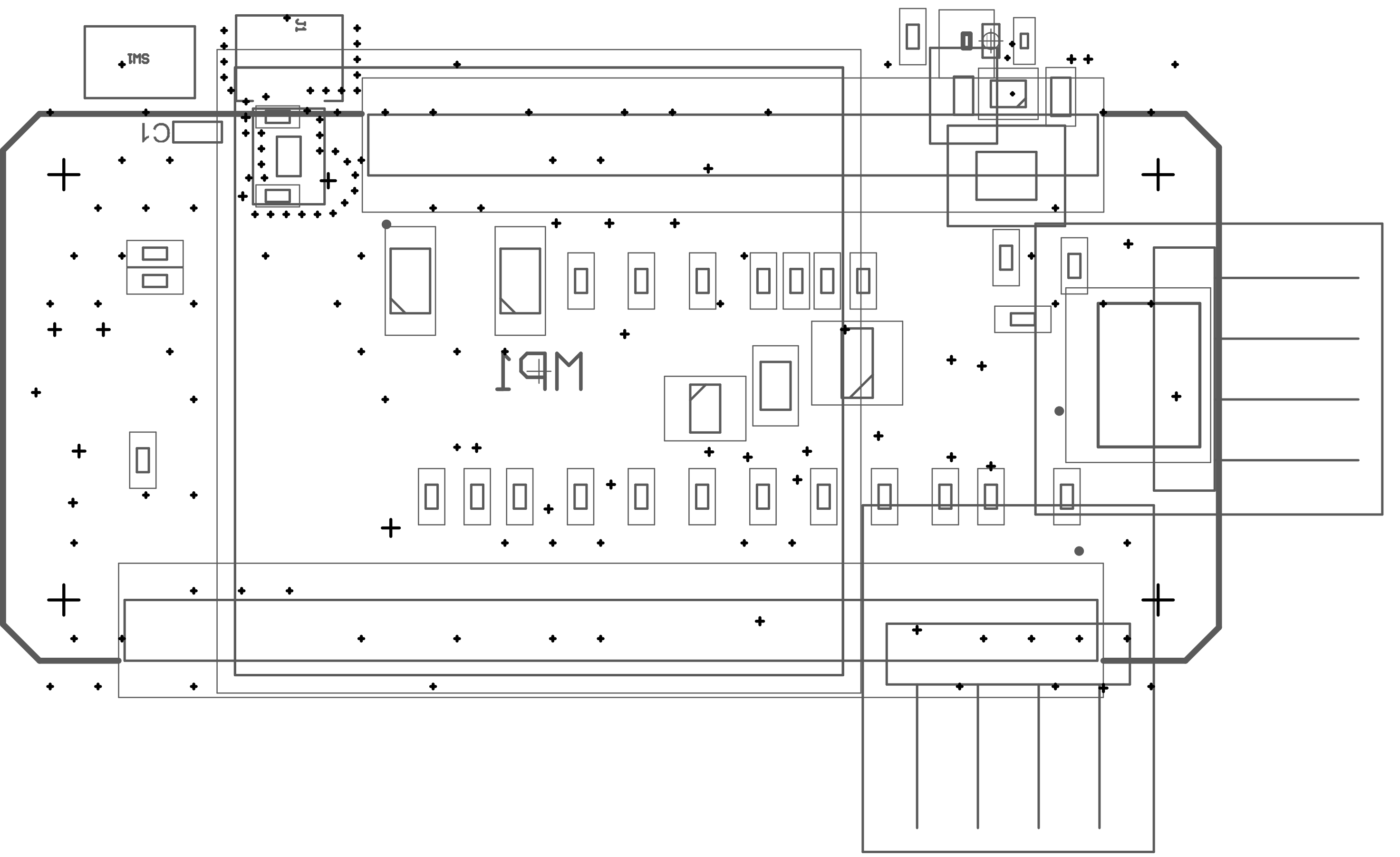
	Bottom Overlay				



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	SM-001	1,00mil	4	
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	Bottom Surface Finish	PbSn	0,79mil		
	Bottom Solder	SM-001	1,00mil	4	
	Bottom Overlay				

Comment	Description	Designator	Footprint	LibRef	Quantity
GRM155R71C104KA88D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 0.1 uF 16 VDC 10% 0402 X7R	C1	0402	GRM155R71C104KA88D	1
CC0402BRNPO9BN3R6	Multilayer Ceramic Capacitors MLCC - SMD/SMT 50 V 3.6pF C0G 0402 Tol 0.1pF	C3	CAPC1005X55N	CC0402BRNPO9BN3R6	1
GRM188R61A106MAALJ	Multilayer Ceramic Capacitors MLCC - SMD/SMT 10 uF 10 VDC 20% 0603 X5R	C4	CAPC1608X90N	GRM188R61A106MAALJ	1
GRM188R61A226ME15D	Multilayer Ceramic Capacitors MLCC - SMD/SMT 22 uF 10 VDC 20% 0603 X5R	C5	GRM18x	GRM188R61A226ME15D	1
MBR120LSFT3G	Schottky Diodes & Rectifiers 1A 20V	D1, D2	SODFL3616X98N	MBR120LSFT3G	2
20279-001E-01	CON IPEX MHF SMT RECEPTACLE RF DC - 6GHZ VSWR: 1.3 max. at 0.1-3GHz; VSWR: 1.5 max. at 3-6GHz	J1	MHF SMT REC	20279-001E-01	1
SSM-112-F-SV-P-TR	Headers & Wire Housings Surface Mount, 12pos, PCB Socket Strips, .100" pitch	J2	SAMTEC_SSM-112-F-SV-P-TR	SSM-112-F-SV-P-TR	1
SSM-116-F-SV-P-TR	Headers & Wire Housings Surface Mount, 16pos, PCB Socket Strips, .100" pitch	J3	SAMTEC_SSM-116-F-SV-P-TR	SSM-116-F-SV-P-TR	1
TC2030-NL	CON PLUG OF NAILS 6 PIN FOOTPRINT NO LEGS	J4	TC2030NL	TC2030-NL	1
M20-8890445R	Connector	J5, J6	M208890445R	M20-8890445R	2
PRT-14417	QWIIC JST CONNECTOR - SMD 4-PIN	J7	SPARKFUN_PRT-14417	PRT-14417	1
LQW18AN18NG8ZD	Inductor Wirewound , 18 nH, 75 Milliohm, ± 2%, 1.4 A, 0603 [1608 Metric]	L1	LQW18AN10NG80D	LQW18AN18NG8ZD	1
DFE252012F-2R2M=P2	Inductor Power Chip Shielded Wirewound 2.2uH 20% 1MHz Metal 2.3A 0.082Ohm DCR 1008 T/R	L2	DFE2HCAH1R0MJ0L	DFE252012F-2R2M=P2	1
DMG3415U-7	MOSFET (P-Channel) 16V Enh FET 8Vgss -12A 0.65W	Q1	SOT96P240X120-3N	DMG3415U-7	1
BSS138BKW,115	MOSFET (N-Channel)	Q2	SOT65P210X110-3N	BSS138BKW,115	1
RC0805JR07100KL	Thick Film Resistors - SMD 100 kOhms 125 mW 0805 5%	R1	RESC2012X60N	RC0805JR07100KL	1
0R	Thick Film Resistors - SMD 0402 0.063 W 0 Ohm Jumper	R5, R6, R17, R18, R24	RESC1005X40N	CRCW04020000Z0ED	5
CRCW020136K5FNED	Thick Film Resistors - SMD 36.5Kohm 1% 200ppm	R25	RESC0603X28N	CRCW020136K5FNED	1
RC0201FR-070RL	Thick Film Resistors - SMD 0 Ohms 50mW 0201 1%	R26	RESC0603X26N	RC0201FR-070RL	1
ERJ-1GNF1622C	Thick Film Resistors - SMD 0201 16.2Kohm 1% HalogenFree AEC-Q200	R27	ERJ1GN_(0201)	ERJ-1GNF1622C	1
CRCW040210K0JNEDC	Thick Film Resistors - SMD 1/16watt 10Kohms 5% Commercial Use	R28, R30	RESC1005X40N	CRCW040210K0JNEDC	2
stm32wl-SMD	Arribada SMD module Argos comunication with PA GR5504	SMD1	stm32wl-SMD_noLegs	stm32wl-SMD	1
PTS820J2x	SW SPST PB 50MA 12V SMT CK variant	SW1	TL1015	PTS820J2x	1
TPS63901YCJR	Integrated Circuit	U1	BGA12C35P3X4_111X146X35	TPS63901YCJR	1